# Nassau Candy Distributor

## Product Line Profitability & Margin Performance Analysis

Step-by-Step Project Execution Workflow

**Purpose:** This document structures the project execution workflow according to the supplied technical documentation, from data cleaning through final business deliverables.

## Phase 1: Data Cleaning & Validation

### *Cost & Sales Audit*

- Verify cost and sales values, removing records with zero sales or invalid profit figures.

### *Missing Value Handling*

- Resolve missing unit values or customer demographic details.

### *Label Standardization*

- Standardize product labels, division names, and factory mapping.

## Phase 2: Metric Calculation & Feature Engineering

Calculate key financial metrics for every order and product line:

| Metric | Formula / Definition |
|---|---|
| Gross Margin (%) | Gross Profit ÷ Sales |
| Profit per Unit | Gross Profit ÷ Units |
| Revenue Contribution | Product Sales ÷ Total Sales |
| Profit Contribution | Product Profit ÷ Total Profit |

## Phase 3: Core Analytical & Diagnostic Modules

### *Product-Level Analysis*

- Group products into performance tiers: High-Profit/High-Margin, High-Sales/Low-Margin (risk zone), and Low-Sales/Low-Profit.

### *Division-Level Performance*

- Compare revenue vs. profit contribution per division to uncover structural margin imbalances.

### *Pareto (80/20) Concentration Analysis*

- Determine the specific product portfolio driving 80% of total revenue and 80% of total profit.

### *Cost Structure Diagnostics*

- Analyze cost-vs-sales relationships to highlight items requiring repricing, cost renegotiation, or discontinuation review.

## Phase 4: Streamlit Web Dashboard Development

Build an interactive multi-tab dashboard (**app.py**) featuring:

### *1. Product Profitability Overview*

• Leaderboard rankings and profit contribution charts.

### *2. Division Performance Dashboard*

• Revenue vs. profit comparison and margin distribution.

### *3. Cost vs. Margin Diagnostics*

• Interactive scatter plots with margin risk flags.

### *4. Profit Concentration Analysis*

• Pareto visualization charts and dependency indicators.

### *5. Interactive Controls*

• Date range picker, division filter, margin threshold slider, and product search bar.

## Phase 5: Final Deliverables

### *Research Paper*

• Detailed report covering exploratory data analysis (EDA), findings, and recommendations.

### *Live Dashboard*

• Production-ready Streamlit web application.

### *Executive Summary*

• High-level strategic briefing tailored for stakeholders.

## End-to-End Workflow

| Step | Project Stage |
|---|---|
| 1 | Data Cleaning & Validation |
| 2 | Metric Calculation & Feature Engineering |
| 3 | Product & Division Analysis |
| 4 | Pareto & Cost Diagnostics |
| 5 | Streamlit Dashboard Development |
| 6 | Research Paper + Live Dashboard + Executive Summary |

**Source basis:** The workflow is structured from the project execution steps provided by the user and aligned with the supplied Nassau Candy technical documentation.